

Simulation Evidence & Reliance Boundary Standard - (SERBS)

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Independent Methodological Authority

Simulation Evidence & Reliance Boundary Standard (SERBS) is the Parent Standard within SIMULOS® — Simulation Governance Architecture that governs what simulation-derived information may legitimately become evidence, what evidential meaning and weight that evidence may carry, how far reliance upon it may extend, which uses must remain limited or prohibited, and how simulation-derived evidence must interface with broader validation governance.

SERBS establishes the methodological boundary between simulation output, simulation-derived evidence, and validation reliance.

It requires simulation evidence to remain connected to the Simulation Object, simulation purpose and scope, represented reality, assumptions, model and environment, input basis, scenario conditions, execution record, governed outputs, and simulation-to-reality correlation from which its evidential meaning originates.

SERBS does not determine whether an object, system, claim, process, or operational proposition is validated. It governs the simulation-specific evidential basis that may legitimately be transferred into broader validation processes.

Core Question

What evidential meaning may simulation results legitimately carry, and where does simulation-specific reliance end before broader validation governance becomes necessary?

Architectural Position

SERBS occupies the eleventh governed space of the SIMULOS® simulation governance lifecycle.

It follows:

Simulation-to-Reality Correlation Standard (SRCS)

and precedes:

Simulation Lifecycle & Resimulation Standard (SLRS).

The preceding standards establish:

what is simulated → why it is simulated → how reality is represented → what

assumptions are used → how the model and environment are constituted → what inputs enter → what scenarios are tested → how execution occurs → what outputs are produced → how those outputs correlate with reality.

SERBS then asks the next methodological question:

What may those governed simulation results legitimately support as evidence?

The purpose of SERBS is therefore not to produce additional simulation results, but to establish the legitimate evidential meaning and reliance boundary of results already produced and governed through the preceding SIMULOS® lifecycle.

Governance Flow Position

Governed Simulation Outputs + Governed Simulation-to-Reality Correlation → Simulation Evidence Qualification → Simulation Evidence Weight → Simulation Reliance Boundary → Simulation Use Limitation → Simulation Validation Interface → Governed Simulation Evidence & Reliance Record → Simulation Lifecycle & Resimulation

Where broader validation is required:

Governed Simulation Evidence & Reliance Record → VALIDOS® Validation Governance

Operational Space

SERBS governs the operational space between simulation-derived information and legitimate reliance upon that information.

Its governed space includes: simulation evidence identification; simulation evidence qualification; evidence-purpose alignment; evidence provenance; simulation-specific evidential weight; conditional evidential meaning; evidence applicability; uncertainty affecting evidential meaning; aggregation of evidence across runs or campaigns; reliance boundaries; unsupported reliance; use limitations; prohibited interpretations; temporal applicability of simulation evidence; evidence transfer into broader validation processes; preservation of simulation-specific qualifications during downstream use.

SERBS transforms simulation-derived information from an unbounded analytical product into an explicitly governed evidential basis.

Operational Scope

SERBS applies whenever simulation results are expected to support or influence a material claim, assessment, comparison, engineering judgment, scientific conclusion, design decision, safety argument, mission decision, operational decision, validation process, deployment decision, or other form of reliance.

It applies across: individual simulation runs; repeated simulations; scenario portfolios; large-scale simulation campaigns; engineering simulations; scientific simulations; physical simulations; digital simulations; hybrid simulations; digital twins; infrastructure simulations; safety simulations; mission simulations; economic simulations; AI simulations; autonomous-system

simulations; future simulation technologies.

SERBS remains technology-neutral.

Its concern is not how sophisticated the simulation technology is, but what evidential meaning its results can legitimately support. Core Governance Distinctions Simulation Output ≠ Simulation Evidence

A simulation output is information produced by simulation execution.

Simulation evidence is simulation-derived information that has been explicitly qualified for a defined evidential purpose.

The existence of an output does not automatically create an evidential basis. Simulation Evidence ≠ Validation Determination

Simulation evidence may contribute materially to validation.

It does not automatically establish validation.

A Validation Determination may require simulation evidence together with empirical, operational, observational, experimental, documentary, or other evidence governed through VALIDOS®. Evidence Availability ≠ Evidence Sufficiency

Large quantities of simulation evidence do not necessarily create sufficient evidence.

A million executions based upon the same inaccurate representation, assumption, model limitation, or missing scenario may reproduce the same weakness one million times. Evidence Weight ≠ Evidence Volume

The amount of simulation data is not equivalent to the strength of its evidential meaning.

Evidence weight depends on the governed relationship between the evidence and the proposition it is intended to support. Correlation ≠ Equivalence

Even strong simulation-to-reality correlation does not establish that the simulation and reality are equivalent.

The evidential meaning of a simulation result remains bounded by the conditions under which the correlation was established. Reliance ≠ Unlimited Generalization

Evidence legitimately applicable to one scenario, configuration, operational condition, time period, or purpose must not automatically be generalized to another.

Simulation Evidence Qualification

Simulation-derived information may be treated as governed simulation evidence only where its evidential basis is sufficiently identifiable.

Qualification should establish, where materially relevant: the simulation-derived information being relied upon; the evidential question or proposition; originating Simulation Object; relevant Simulation Object version and configuration; simulation purpose and scope; applicable exclusions;

reality representation; material assumptions; model and environment; input and data basis; relevant scenario or scenario set; execution basis; output identity and classification; inference basis; simulation-to-reality correlation; material uncertainty; temporal applicability; known limitations.

Evidence qualification therefore creates an explicit relationship between:

Evidence → Simulation Basis → Evidential Meaning.

Simulation Evidence Weight

SERBS requires the evidential weight of simulation-derived evidence to remain distinct from the mere existence of the evidence.

Evidence weight may be influenced by: strength of reality representation; representational fidelity; model applicability; assumption dependence; input representativeness; scenario relevance; scenario coverage; execution consistency; repetition results; output classification; inference distance; simulation-to-reality correlation; uncertainty; known limitations; temporal relevance; dependence between evidence sources.

Evidence weight must not be inferred merely from: computational scale; simulation sophistication; graphical realism; institutional prestige; number of runs; number of generated outputs; apparent precision.

Precision of simulation output does not itself establish strength of simulation evidence.

Simulation Evidence Aggregation

SERBS recognizes that simulation campaigns may create evidence across thousands, millions, or more individual executions.

Evidence aggregation must therefore distinguish between:

More executions

and

More independent evidential support.

Repeated executions may strengthen evidence where repetition legitimately tests variability, uncertainty, stochastic behavior, scenario diversity, or reproducibility.

They may add little evidential strength where all executions share the same material: assumption; model weakness; representation error; input deficiency; scenario omission; correlation limitation.

Common-mode weaknesses must therefore remain visible when simulation evidence is aggregated.

Simulation Reliance Boundary

The Simulation Reliance Boundary establishes the maximum simulation-specific reliance that may legitimately be placed upon qualified evidence.

A reliance boundary may be constrained by: Simulation Object; object version; object configuration; purpose; scope; exclusion; represented reality; abstraction; fidelity; assumption; parameter range; model applicability; environmental configuration; input basis; scenario; execution condition; output class; inference; correlation; uncertainty; temporal validity.

Reliance outside these boundaries requires a new governed basis.

SERBS therefore prevents the transformation:

Evidence valid under Condition A → claim assumed valid everywhere.

Simulation Use Limitation

Every material limitation affecting legitimate use of simulation-derived evidence must remain attached to that evidence.

Use limitations may establish that evidence: supports only a defined proposition; applies only to specified scenarios; applies only to a defined Simulation Object configuration; applies only within a defined parameter range; depends upon identified assumptions; excludes particular conditions; does not represent specified real-world phenomena; cannot support specified decisions; cannot independently establish operational performance; requires additional evidence; cannot be generalized beyond a stated environment; expires or requires review after material change; may support comparison but not absolute prediction; may support exploration but not validation; may support validation only together with additional evidence.

A limitation that is separated from the evidence it constrains is no longer an effective governance control.

Unsupported Reliance

SERBS requires unsupported reliance to be identifiable.

Unsupported reliance may arise where simulation evidence is used: outside its simulation purpose; beyond its scope; for an excluded condition; for a different Simulation Object version; beyond tested parameter ranges; outside represented reality; outside scenario coverage; beyond established correlation; beyond temporal applicability; after material change; contrary to a documented limitation; as sole validation evidence where broader evidence is required.

Unsupported reliance must not become legitimate merely because it becomes organizational practice.

Negative Evidence Principle

SERBS recognizes that failed, anomalous, contradictory, unstable, divergent, or unexpected simulation results may carry significant evidential meaning.

Simulation campaigns must therefore not treat only favorable or successful

outcomes as evidence.

Governable simulation evidence may include: failure; non-convergence; instability; unexpected behavior; threshold violation; scenario failure; sensitivity; disagreement between models; divergence from observed reality; unexpected variance; evidence against an intended claim.

An unwanted simulation result does not cease to be evidentially relevant because it is inconvenient.

Temporal Evidence Principle

Simulation evidence has a temporal dimension.

Evidence qualified at one point in time may become less relevant where: the Simulation Object changes; operating conditions change; reality changes; new observations become available; assumptions become obsolete; models are replaced; inputs change materially; new scenarios become relevant; correlation deteriorates.

SERBS therefore connects directly to Simulation Lifecycle & Resimulation Standard (SLRS).

Evidence whose simulation basis has materially changed must not retain an unchanged reliance status automatically.

Validation Interface Principle

SERBS establishes the methodological interface between SIMULOS® and VALIDOS®.

The boundary is explicit:

SIMULOS® determines what simulation-derived evidence can legitimately mean.

VALIDOS® determines whether that evidence is sufficient, alone or together with other evidence, to support a Validation Determination or intended reliance.

SERBS may therefore qualify: what evidence exists; what proposition it supports; what weight it may carry; under what conditions it applies; what uncertainties affect it; what limitations govern its use.

SERBS does not decide: whether the complete Validation Object is valid; whether the complete evidence basis is sufficient; whether validation criteria have been satisfied; whether a Validation Determination may be issued; whether operational reliance is justified by the total validation basis.

Cross-Architecture Boundaries SERBS ↔ VALIDOS®

SIMULOS® / SERBS governs simulation-derived evidence and the legitimate boundary of simulation-specific reliance.

VALIDOS® governs the sufficiency of the complete validation basis and formal Validation Determination.

Simulation evidence qualification ≠ validation sufficiency. SERBS ↔ INTEGROS®

SERBS governs the simulation-specific meaning and use of simulation-derived evidence.

INTEGROS® governs broader integrity properties affecting objects, processes, states, information, and evidence.

SERBS must therefore not redefine general evidence integrity. SERBS ↔ ORA™

ORA™ governs actual Operational Reality.

SIMULOS® may use governed Operational Reality and SRCS correlation relationships when establishing simulation evidential meaning.

SERBS does not determine what operational reality is. SERBS ↔ ADIT®

ADIT® may govern auditability, preservation, evidential observation, verification, and examination of what occurred.

SERBS governs the simulation-specific meaning and reliance boundary of simulation-derived evidence.

Auditability of evidence ≠ evidential meaning of simulation evidence. SERBS ↔ AGA™

AGA™ governs who holds authority, responsibility, accountability, and attribution for decisions involving simulation evidence.

SERBS governs what the evidence can legitimately support. SERBS ↔ AIG® / ASGA™

SIMULOS® may generate evidence concerning AI or autonomous systems.

SERBS governs that simulation-derived evidence.

AIG® and ASGA™ continue to govern the actual behavior, authority, autonomy, constraints, and operation of the systems themselves. Parent Standard Modules
1. SEQM — Simulation Evidence Qualification Module

Governs the transformation of simulation-derived outputs, observations, correlations, and inferences into explicitly qualified simulation evidence.

SEQM establishes: evidence identity; evidential purpose; simulation basis; provenance; qualification conditions; applicability; uncertainty; evidential qualification state.

Its core question is:

What simulation-derived information legitimately qualifies as evidence, and for what evidential purpose? 2. SEWM — Simulation Evidence Weight Module

Governs the simulation-specific evidential weight assigned to qualified simulation evidence.

SEWM evaluates factors that materially strengthen, weaken, condition, or constrain evidential meaning.

Its core question is:

How much evidential significance may qualified simulation evidence legitimately carry within its defined context? 3. SRBM — Simulation Reliance Boundary Module

Governs the maximum legitimate reliance that may be placed upon simulation-derived evidence within the simulation governance context.

Its core question is:

How far may reliance upon this simulation evidence legitimately extend before the evidence is being used beyond its governed basis? 4. SUM — Simulation Use Limitation Module

Governs conditions, exclusions, restrictions, caveats, unsupported interpretations, and prohibited uses attached to simulation evidence.

Its core question is:

What must simulation-derived evidence not be used to claim, infer, justify, or support? 5. SVIM — Simulation Validation Interface Module

Governs the transfer boundary between SIMULOS® simulation evidence governance and VALIDOS® validation governance.

SVIM ensures that evidence entering validation retains its: identity; provenance; qualification; evidential weight; applicability; uncertainty; reliance boundary; limitations.

Its core question is:

How must simulation-derived evidence enter broader validation governance without losing the conditions that define its legitimate meaning?

Internal Module Flow

Governed Simulation Outputs + Governed Simulation-to-Reality Correlation → SEQM — Simulation Evidence Qualification Module → SEWM — Simulation Evidence Weight Module → SRBM — Simulation Reliance Boundary Module → SUM — Simulation Use Limitation Module → SVIM — Simulation Validation Interface Module → Governed Simulation Evidence & Reliance Record

The internal logic is:

What qualifies as evidence → what weight it carries → how far reliance may extend → what uses remain prohibited or constrained → how evidence may enter broader validation governance.

Minimum Implementation Framework

A minimum SERBS implementation should establish: 1. Evidence Object

Identify the precise simulation-derived information proposed for evidential use. 2. Evidential Purpose

Define the proposition, question, decision, or claim the evidence is intended to inform. 3. Simulation Basis

Trace the evidence to its originating Simulation Object, purpose, scope, representation, assumptions, model, inputs, scenarios, executions, outputs, and correlation basis where relevant. 4. Evidence Qualification

Determine whether the simulation-derived information legitimately qualifies as evidence for the stated purpose. 5. Evidence Weight

Determine the simulation-specific weight that the evidence may legitimately carry. 6. Reliance Boundary

Establish the maximum context in which reliance remains supported. 7. Use Limitations

Record material exclusions, restrictions, unsupported interpretations, and prohibited uses. 8. Uncertainty

Preserve uncertainty affecting evidential meaning. 9. Validation Interface

Preserve all relevant evidence conditions when evidence enters VALIDOS® or another downstream validation process. 10. Lifecycle Sensitivity

Identify changes capable of invalidating or weakening the current evidential basis and triggering review or resimulation.

Governance Outputs

SERBS may generate: Simulation Evidence Qualification Record Simulation Evidence Basis Record Simulation Evidence Weight Assessment Simulation Evidence Provenance Record Simulation Evidence Applicability Statement Simulation Reliance Boundary Statement Simulation Use Limitation Record Unsupported Reliance Register Simulation Evidence Uncertainty Record Simulation Evidence Aggregation Record Simulation Validation Interface Package Governed Simulation Evidence & Reliance Record

Together, these outputs make the evidential meaning of simulation results explicit, transferable, reviewable, and bounded.

Enterprise Application

SERBS enables an organization to determine not merely whether a simulation has produced results, but what those results can legitimately support.

It can be used to govern simulation evidence used for: engineering decisions; design decisions; safety cases; scientific claims; infrastructure planning; mission preparation; product development; system qualification; operational planning; AI evaluation; autonomous-system evaluation; validation programs; regulatory submissions; investment decisions; risk analysis; scenario comparison.

SERBS is particularly relevant where simulation produces highly convincing, precise, or large-scale outputs that may create an appearance of certainty greater than their actual evidential basis.

A simulation may produce billions of data points without adequately supporting the real-world proposition for which those data are being used.

Equally, a narrowly designed simulation may provide strong evidence for a precisely bounded question even though it cannot support broader claims.

SERBS gives organizations the governance structure needed to distinguish between these situations.

Foundational Principles

Simulation output is not automatically simulation evidence.

Simulation evidence is not automatically validation evidence sufficient for a Validation Determination.

Evidence availability is not evidence sufficiency.

Evidence volume is not evidential strength.

Repeated simulation is not automatically independent evidence.

Simulation complexity is not evidential authority.

Numerical precision is not evidential certainty.

Correlation with reality does not establish equivalence with reality.

Evidence must remain bounded by the conditions that give it meaning.

Use limitations are part of the evidence itself.

Negative and anomalous simulation results may carry material evidential meaning.

Simulation-derived evidence must not be relied upon beyond the object, conditions, scenarios, assumptions, executions, outputs, correlations, and temporal context from which its evidential meaning was established.

SIMULOS® determines what simulation-derived evidence can legitimately mean. VALIDOS® determines whether the total evidence basis is sufficient to support a Validation Determination or intended reliance.

Canonical Closing Statement

Simulation Evidence & Reliance Boundary Standard establishes the evidential boundary layer of SIMULOS® — Simulation Governance Architecture by transforming simulation-derived outputs, observations, correlations, and inferences into explicitly qualified, weighted, bounded, limitation-aware, and validation-interface-ready simulation evidence. Through governance of evidential purpose, qualification, provenance, weight, applicability, uncertainty, aggregation, reliance boundaries, use limitations, unsupported

interpretations, temporal relevance, and validation transfer, SERBS prevents simulation results from acquiring greater authority, certainty, generality, or operational meaning than their governed simulation basis can legitimately support. By separating simulation evidence qualification from broader validation sufficiency, SERBS preserves the methodological boundary between SIMULOS® and VALIDOS® while enabling simulation-derived evidence to enter validation, engineering, scientific, safety, mission, operational, and decision processes without losing the conditions that define its legitimate meaning. SERBS therefore establishes the final evidential gateway of the active simulation process before lifecycle governance begins, ensuring that no simulation-derived evidence can legitimately be relied upon more broadly than the Simulation Object, purpose, reality representation, assumptions, model, inputs, scenarios, executions, outputs, correlations, uncertainties, limitations, and temporal conditions from which that evidence was derived.

Module Architecture

→ [Simulation Evidence Qualification Module \(SEQM\)](#)

→ [Simulation Evidence Weight Module \(SEWM\)](#)

→ [Simulation Reliance Boundary Module \(SRBM\)](#)

→ [Simulation Use Limitation Module \(SUM\)](#)

→ [Simulation Validation Interface Module \(SVIM\)](#)

Related Documents

→ [About Simulation Evidence & Reliance Boundary Standard \(SERBS\)](#)

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